

NETWORK THREAT DETECTION WITH MACHINE LEARNING

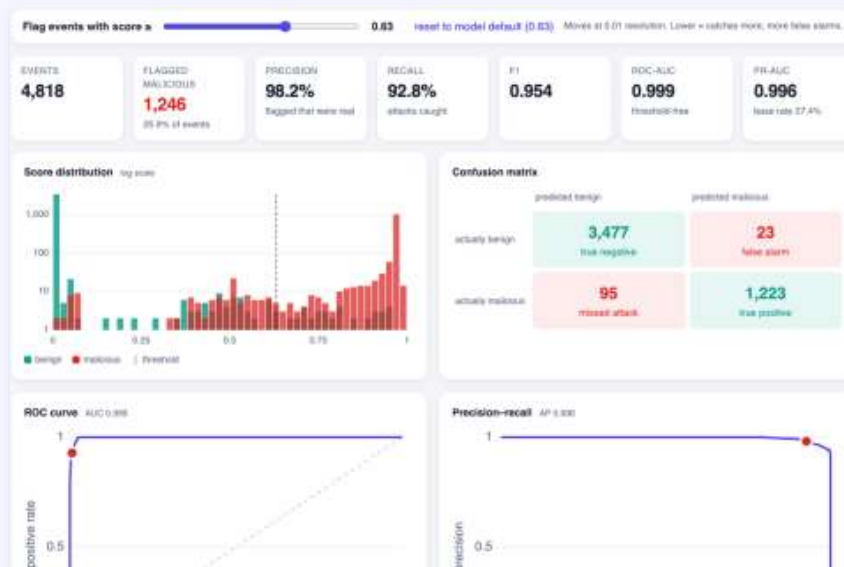
SIEM ML Predictor

Upload network events. A pre-trained model scores every one as malicious or benign, and shows why.

UNSW-NB15

XGBoost

FastAPI



Live app: [full statistics view](#)

The problem: analysts drown in events

- SIEM tools export millions of network events every day
- Most events are benign, but every one needs a look
- Manual rules miss new patterns and raise false alarms
- Slow triage gives attackers more time inside

What the model does to a real capture

1,140,045

network flows in the test capture

224,530

flagged for review · 19.7%

218,787

real attacks · 97.4%

The solution: upload, score, understand



1. Upload

- Drop a CSV, Parquet, JSON or NDJSON file
- Up to 300 MB or 2M rows
- Missing columns are reported, not fatal



2. Score

- Pre-trained XGBoost model
- Every event gets a 0 to 1 attack score
- Flagged above the tuned threshold (0.63)

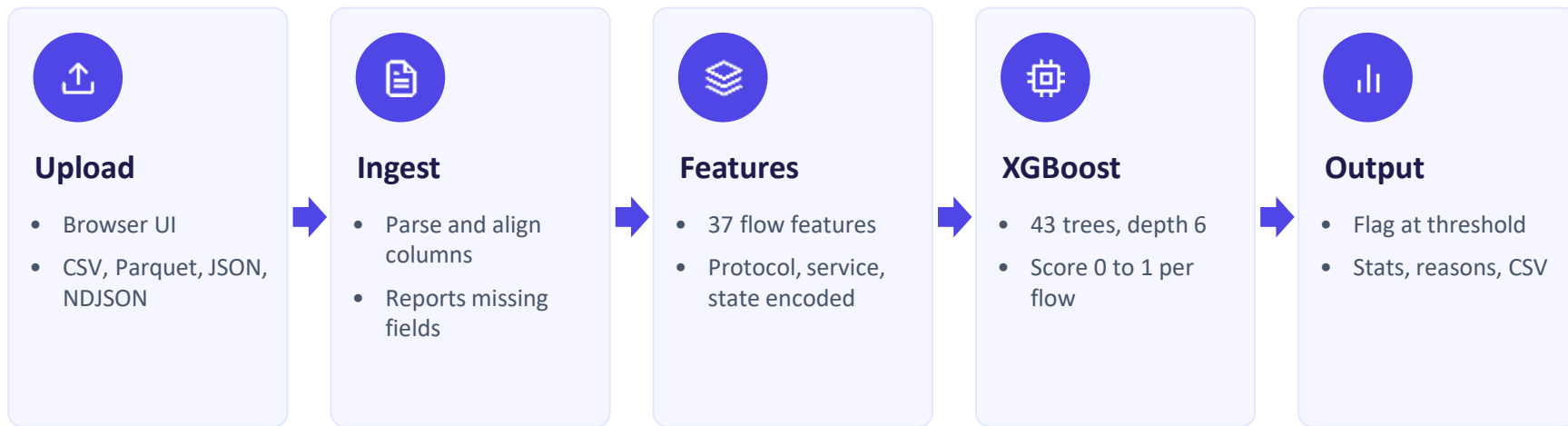


3. Understand

- Precision, recall, F1, ROC-AUC
- Confusion matrix and per-attack detection
- Plain-word reasons behind each flag

No accounts. No database. No case management. Just a file in and statistics out.

How it works: from file to flagged events



Tech stack



Data and model behind the scores

912K

training flows

1.14M

held-out test flows

37

flow features

43

trees, depth 6

- Dataset: UNSW-NB15 raw network flows from UNSW Canberra
- Trained on an early capture, threshold tuned on a later slice
- Tested on a separate later capture the model never saw
- IPs, client ports and timestamps left out: they identify the lab, not the behaviour
- Threshold 0.63 maximises F1 on validation

Results on a capture the model never saw

97.4%

Precision

96.2%

Recall

0.968

F1 score

0.999

ROC-AUC

0.63%

False alarms

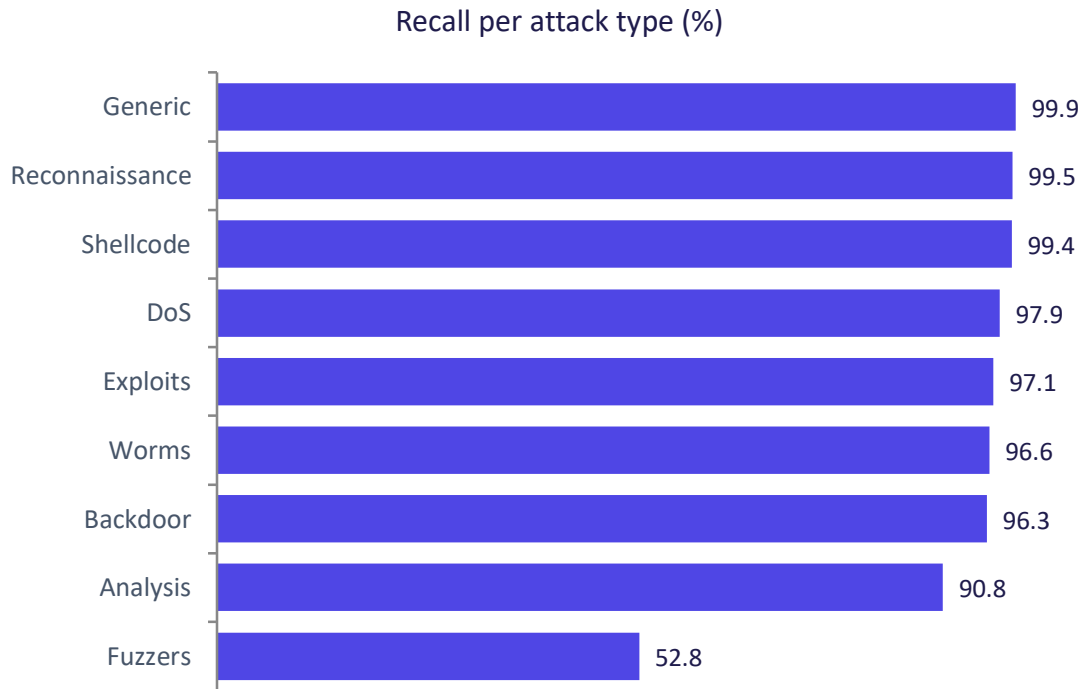
Confusion matrix (1,140,045 flows)

	Predicted benign	Predicted malicious
Actually benign	906,858 true negative	5,743 false alarm
Actually malicious	8,657 missed attack	218,787 true positive

What it means

- Of 100 flagged events, about 97 are real attacks
- Of 100 real attacks, about 96 are caught
- Only 0.63% of benign traffic raises a false alarm
- ROC-AUC 0.999: attacks are ranked far above normal traffic

Detection by attack type



Strong

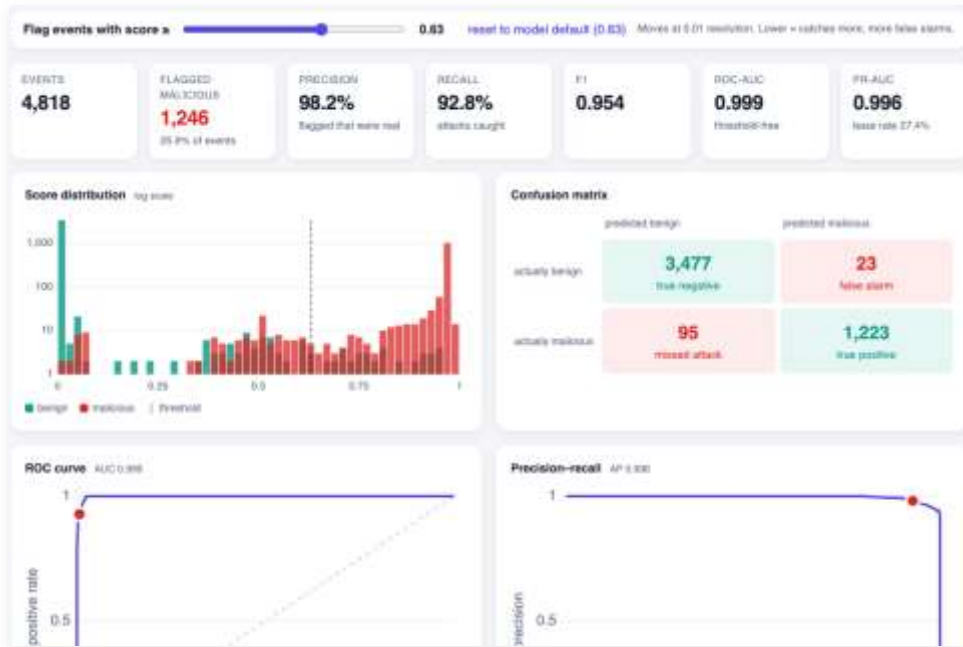
Seven of nine types above 96% recall. Generic, the largest class, reaches 99.9% (151K flows).



Weak spot: Fuzzers

Only 52.8% recall: 7,814 of 14,806 caught. Fuzzing traffic is the hardest to separate from normal.

Product tour: see how every score is made



Full statistics: threshold slider, KPIs, score distribution, confusion matrix



Why flagged: per-event reasons

- 9-step animated walkthrough of every scoring stage
- Live threshold slider recomputes all metrics
- Reasons in plain words for each flagged event
- One-click CSV of all predictions

Priority tiers and IBM Bob integration

Priority tiers for triage

Critical

Score 0.90 to 1.00 · 214,189 flows
99.2% real · holds 93.4% of attacks

High

Score 0.63 to 0.90 · 10,341 flows
61.1% real · holds 2.8% of attacks

Watch

Score 0.30 to 0.63 · 21,876 flows
34.8% real · holds 3.3% of attacks

Low

Score below 0.30 · 893,639 flows
0.1% real · holds 0.5% of attacks

IBM Bob via MCP

Same code as the web app, exposed as four read-only tools:

- `get_model_info`: which model, how far to trust it
- `list_samples`: sample flow files to try
- `analyze_file`: which flows look malicious, what to check first
- `explain_event`: why the model flagged row N

Ask Bob: "Which flows in sample_labelled.csv look malicious, and what should I check first?"

Limits, next steps and takeaway



Honest limits

- UNSW-NB15 is a 2015 lab testbed: expect lower scores on a real network
- Source TTL alone reaches ROC-AUC 0.983, so the data is easy
- Fuzzers only 52.8% recall
- Reads flow records only, not syslog or CEF text
- Scores rank events for review, not a verdict



Next steps

- Train and validate on real enterprise flow logs
- Add a text model for syslog and CEF events
- Fuzzer-specific features to lift recall
- Analyst feedback loop for periodic retraining

Fast, explainable, honest triage help. Thank you. Questions?